Monthly Report for ECE FYP/FYT

Project Code:	YH02	Supervisor(s):	Yin Huan
Project Title:	Development of a Multi-Camera Vision-Based Localization System for Tracking Multiple Micro Autonomous Underwater Vehicles (AUVs)		
Group Member(s):	1) CHAN, Sze Sen 2) XU, Borong		
Reporting Period:	Report #1 ☑ Oct (Fall) Report #2 ☐ Nov (Fall) Report #3 ☐ Feb (Spring) (please attach Reports #1-2 to the Progress Report to be submitted in Jan) (please attach Reports #3 to the Final Report to be submitted in Apr)		
Progress Report: ● List the work completed in this reporting period. ● Identify the major difficulties encountered. ● Comment on the overall progress.	Work completed: ● Calibrated the camera's intrinsic parameter for single-camera localization in the air. ● Designed the camera array and mounted it under the experimental water tank. ● Designed a new method for camera calibration using ChArUco boards to improve the accuracy of cross-camera views. Major difficulties: ● The theoretical model of multi-camera calibration is yet to be confirmed, requiring more literature review. ● The gap between an in-air environment and an underwater cross-medium environment is yet to be confirmed. Underwater experiments need to be conducted. Overall progress: ● Current progress is matching our goals.		
Future Plan: ● Write down the working plan for the next reporting period.	● Discover the implementation of RSSI as an assisting localization method through literature reviews. ● Model the cross-medium light refraction and develop an algorithm to fit the distortion. ● Conduct underwater experiments.		
Group Representative's Signature:			

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